Chapter 3: Predictive Analytics – Decision Trees

Multiple choice questions

Question 1

Level: medium

Which statement is **TRUE**?

a) Estimating class probabilities in the leaf nodes of a classification tree is a better approach than majority voting for making the splitting decision in learning a decision tree, since it allows to differentiate among the instances that are classified in the same class.
b) In addition to learning interaction effects, decision trees can easily mimic linear relations or decision boundaries; therefore, they typically outperform linear methods.
c) Decision trees can be transformed into a set of decision rules by following the various paths from the root node to the leaf nodes.
d) The information gain criterion allows to select the split that maximizes the decrease in impurity, by comparing the impurity of the parent node and the average impurity of the child nodes.

Question 2

Level: easy

Which statement is **FALSE**?

a) Decision tree learning methods such as C4.5 and CART adopt a greedy approach for making the splitting decision and therefore may not learn the globally optimal tree for the training data.
b) Decision trees can learn and incorporate both linear relations and interaction effects.
c) Regression trees can both minimize the error within leaf nodes and maximize the difference between predicted values of leaf nodes.
d) A greedy approach is adopted for learning decision trees, because the number of possible trees grows exponentially with the number of variables in the dataset.

Open questions

Question 1

Which three types of decisions are to be made by decision tree learners?

Question 2

Explain the use of the gini measure, majority voting and tuning in learning a decision tree.

Question 3

What is overfiting? How is overfitting avoided by early-stopping?

Question 4

Explain the steps in the ID3 decision tree algorithm with standard deviation reduction.